

Environmental and Climate Justice in Computing

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ABSTRACT

While climate change has been a longstanding concern of HCI and CSCW communities, this scholarship has rarely drawn attention to the well-documented pattern of minoritized and marginalized communities unfairly carrying the brunt of environmental burdens. Through this one-day remote workshop, we plan to critically extend how CSCW can support climate action by focusing on two social movements, environmental and climate justice, both of which aim to reduce environmental degradation and pursue sustainable communities without doing so at the expense of others. In this workshop, we aim to identify how CSCW and datafication have helped to uphold environmental or climate justice commitments or has been complicit in producing or maintaining environmental harms. We also plan to discuss and identify a CSCW research agenda addressing how to support climate justice principles and processes in designing technologies and systems. We hope that this workshop will help to initiate and foster a longer-term relationship with researchers, activists and practitioners who are engaging with or interested in climate justice in computing.

CCS CONCEPTS

• **Human-centered computing** → **Collaborative and social computing**; **Collaborative and social computing systems and tools**; **HCI theory, concepts and models**;

KEYWORDS

climate change, environmental justice, climate justice, sustainable HCI, digital inequality

ACM Reference Format:

Olivia Doggett, Jen Liu, Ufuoma Ovienmhada, Samar Sabie, Sarah Gram, Laura J Perovich, Matt Ratto, and Robert Soden. 2023. Environmental and Climate Justice in Computing. In *Computer Supported Cooperative Work and Social Computing (CSCW '23 Companion)*, October 14–18, 2023, Minneapolis, MN, USA. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3584931.3611296>

1 BACKGROUND AND MOTIVATION

Over the past 20 years, sustainability related HCI and CSCW scholarship has transformed in response to evolving social, political, and ecological interpretations of how technologies and systems can understand and respond to climate change. Shifting from an individualistic focus on environmental responsibilities and the persuasive technologies that accompany this viewpoint, many contemporary HCI scholars are now committed to systems-level investigations of climate change with a wider range of stakeholders [13, 16, 26, 28]. Many scholars also adopt a more-than-human approach to expand our perspectives on climate challenges [19]. In this workshop, we build upon these critical extensions of how HCI can support climate action by interrogating why and how CSCW is connected to social justice movements that are dually focused on environmental sustainability.

We ground our workshop in two specific movements: environmental and climate justice. Spearheaded primarily by Black activists in the 1980s [18], the environmental justice movement calls attention to the well-documented pattern of minoritized communities unfairly carrying the brunt of environmental burdens (such as air pollution) and unequal access to environmental benefits (such as green space) [1, 20, 22, 23, 27]. The movement thus aims to reduce and redistribute environmental burdens and benefits in a more equitable manner. Similarly, climate justice calls for urgent actions to mitigate climate change effects, particularly through inclusive, community-driven practices [31]. Closely related to the climate justice movement is just transition, an approach recognizing that “equity and justice need to be accounted for during a transition” [8] to define “what is just and for whom” [21] particularly for those

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CSCW '23 Companion, October 14–18, 2023, Minneapolis, MN, USA

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ACM ISBN 979-8-4007-0129-0/23/10.

<https://doi.org/10.1145/3584931.3611296>

who are impacted by green transitions in their jobs, homes, and communities [9].

Together, environmental and climate justice aim to reduce environmental degradation and pursue sustainable communities without doing so at the expense of others. These objectives are critical to connect to HCI scholarship as more traditional conceptions of sustainability, particularly those focused on individual energy savings, may have differential impacts to groups across categories of social difference. For example, smart meters that charge residents a higher rate when grid intensity is high can be detrimental for low-income earners and those who work non-standard 9 - 5 jobs [11]. By centering environmental and climate justice as a core part of climate work within CSCW, we aim to shift the focus from attending to environmental concerns to addressing questions of uneven distributions of environmental harms and benefits.

Emerging work in CSCW and HCI has already begun to confront these imbalances. These works problematize traditional conceptions of sustainability in HCI, which may not be as inclusive of environmental harms related to race, class, and other social categories of difference. For example, Chidisiwasno et al. [2]'s work on designing for power blackouts in Kenyan households and Liu's [15] research on failing network infrastructures in Louisiana illustrate the need to connect our computing infrastructures to our fossil fuel dependencies and the corresponding issues that arise related to climate change and energy access. Other recent work addressing the future of our food systems [3, 4, 17] emphasize the need to focus on marginalized and racialized histories that relate to farming, labor and land distribution as these narratives have been neglected in agricultural technological development and design.

Pushing the boundaries of research in CSCW beyond sustainability is critical as environmental injustices can be structurally fragmented, intentionally made illegible, and distributed over a long time, making them difficult to track [14, 22, 32]. Given CSCW's deep knowledge of language and concepts to make invisible aspects of technologies, labor, and infrastructural systems visible, the discipline is well-resourced ideologically and methodologically to contribute to environmental and climate justice research and design work [29, 30]. Addressing environmental injustice is not only about visibility, however, as many harms are created, regulated, and perpetuated by the state [24]. CSCW's focus not only on visibility but on promoting interventions to change the status quo can extend environmental and climate justice toward more action-oriented outcomes. For example, ongoing CSCW work related to labor justice focuses on ways to support workers through increased participation in research and design, deeper recognition of expertise, and fairer resource distribution [5–7, 10, 12, 33].

To tie environmental and climate justice objectives to CSCW, we anchor our workshop discussion in a central question: **as individual researchers, practitioners, and activists, in what ways does our work, and the research-related institutions we operate within, work against and toward environmental and climate justice movements?** We address this question in two ways. First, by reflecting together at an individual and institutional level to consider how practices and methods of academic research (e.g., extractive data collection, unsustainable paper production cycles) support or counter environmental and climate justice objectives. Second, by co-creating a research agenda addressing what

research needs to be undertaken to align with environmental and climate justice work. We invite researchers, practitioners and activists who are engaged or interested in pursuing environmental and climate justice-related work in the CSCW space. This work includes but is not limited to such areas as critical data studies, computing networks, future of work, social movements, more-than-human relations, and participatory design.

2 WORKSHOP THEMES

In this workshop, we invite contributions on the following themes:

- (1) **Datafication Imaginaries:** Reflections or analysis on the ways data and technology are designed, structured, implemented and disseminated to work toward or against environmental or climate justice outcomes;
- (2) **Visibility and Beyond:** Analysis on the role and relevance of visibility as intervention, as well as the potential for CSCW to support more radical interventions (e.g., land back, prison abolition) as an approach to environmental and climate justice outcomes;
- (3) **Towards Critical Environmental Justice [23]:** Identification of communities (including more than humans) and types of social difference that have been marginalized by traditional notions of sustainability in CSCW;
- (4) **Methodologies:** Opportunities or limitations of existing methodological practices in CSCW (e.g., participatory design, co-production etc.) or academic norms (data collection, paper publication) towards advancing environmental and climate justice.

3 WORKSHOP OUTCOMES

Our proposed one-day remote workshop outcomes are the following:

- (1) Identify how HCI, CSCW and/or datafication has helped to uphold environmental or climate justice commitments or been complicit in producing or maintaining environmental harms and unjust transitions;
- (2) Cultivate a community of researchers, educators, and designers interested in topics related to climate action, environmental and climate justice, repair and maintenance, and critical making [25] as well as unmaking [26];
- (3) Discuss and identify a CSCW research agenda addressing how we should support environmental and climate justice principles and processes in designing technologies and systems.

4 PRE-WORKSHOP LOGISTICS

4.1 Key Dates

- (1) Call for participation released on website: July 6, 2023
- (2) Participant submissions due: August 16, 2023
- (3) Notify participants of acceptance: August 24, 2023
- (4) Assign participants feedback and reflection roles: September 12, 2023
- (5) Workshop date: October 14, 2023

4.2 Website

We will set up a website, which will include workshop description, call for submissions, and application. We will post the papers on the website at least 6 weeks prior to the beginning of the conference.

4.3 Workshop Recruitment

We seek approximately 15-20 participants who are interested or currently engaged in research or design that leverages environmental or climate justice. We will promote our workshop through our website, CHI and CSCW-related mailing lists, and co-organizers' professional social media channels and personal contacts.

4.4 Workshop Application

Participants will be expected to submit a 2–4-page research or position paper, a creative contribution, or an example of political, technological, or design intervention or campaign they are working on. If participants wish to submit a creative contribution or intervention, we ask them to submit a 300–500-word description of their work. We expect submissions to address one or more of our workshop themes. Participants will submit their work via a short online application published on our workshop website.

4.5 Pre-Workshop Participant Engagement and Preparation

Once all contributions have been finalized and accepted, we will circulate them with accepted participants. We expect participants to read the contributions prior to the workshop. We will also invite participants to Discord, Miro, and a Zotero Library. We will encourage participants to introduce themselves in the Discord channel and use it to communicate for feedback and collaboration prior to the workshop. The Miro board will be created by co-organizers about a month prior to the workshop, and we will invite participants to provide feedback. The Zotero Library will consist of all submissions' bibliographies. This library will be a first attempt at a shared resource of environmental and climate justice-related literature for the CSCW community.

5 WORKSHOP STRUCTURE

5.1 Workshop Set-Up

The workshop will occur synchronously for participants, accommodating time differences wherever possible. We will host the session online via Zoom. Workshop materials will be available on our website. To coordinate our remote workshop, our organizers will monitor the Miro board and online chat, contacting in-person organizers via a back-end channel when needed. Organizers will also oversee logistics such as timekeeping, note-taking, and documenting zine materials. We will ask for participant permission before documenting.

5.2 Activities and Timing

Our workshop will take place over the course of 8 hours (including 1.5 hours for breaks). The main activities in our workshop include introductions, defining environmental and climate justice, lightning talks, a mapping activity, a zine-making activity, and a final share-out (see Table 1 for the full schedule).

Table 1: One-Day Remote Workshop Schedule

Time	Activity
9:00-9:30	Introductions
9:30-10:00	What is Environmental and Climate Justice?
10:00-11:15	Lightning Talks and Small Group Discussions
11:15-11:30	Break
11:30-12:30	Small Group Mapping Activity and Share-Out
12:30-3:30	Lunch
13:30-15:15	Zine-making Activity
15:15-15:30	Break
15:30-16:30	Share-out and Discussion
16:30-17:00	Reflection and Closing

We will begin the workshop with introductions and a brief activity discussing environmental and climate justice. Through this activity, we aim for participants to contextualize their work in the broader goals and timelines of these movements. We describe the remaining workshop activities below.

Lightning talks: Our first session will be brief lightning talks from each participant. In these talks, we ask participants to share a high-level overview of their workshop submission, including their area of study, research aims, community-collaborations, methodologies, and open questions and concerns regarding environmental and climate justice in their work. We will ask participants to write any questions or ideas that they generate from listening to the presentations in the Miro board.

Mapping Activity: The goal of this activity is for participants to be able to contextualize their work within the broader goals and timelines of the environmental and climate justice movements. In the first part of the activity, organizers will give an overview of these movements and situate this work within a longer history of activism and collective organizing. In this overview, commitments and principles of the environmental and climate justice movement will be shared. These materials will be made available via Miro and also as hard copies in the on-site room.

In the second part of the activity, participants will then work in small break-out groups of approximately 2 - 3 people based on topics related to their research. These break-out groups will be created based on the four workshop themes. During this time, break-out groups will discuss their research goals and motivations in connection with environmental and climate justice commitments and principles, including how their research may align or work against either movement. Participants are then tasked with adding, revising, or extending these principles. Each group will be asked to share a brief summary of their discussion.

Zine-making activity: For this activity, we will collaboratively create a zine that suggests an initial agenda for how research in CSCW and HCI can be aligned with environmental and climate justice. Based on conversations and notes in the earlier part of the workshop, organizers and participants will collectively brainstorm what could be included as part of a research agenda. From there, workstations will be created based on these proposed research agenda. For the scope of the workshop, we aim to have between 4 - 6 agenda items. In groups of 3 - 4, participants will have approximately 20

minutes to engage with and contribute to each workstation, which will be set up in Miro. Following a short break, we will review all the zine station contributions, and reflect together at a higher level of shared questions, goals, and challenges that run between them. This discussion will inform an informal collective research agenda, which we will also share in the zine. We will close the day reflecting on our workshop experiences and discussing next steps.

6 POST-WORKSHOP PLAN

Following the workshop, we plan to publish our zine via our website. Based on participant consent, we hope to make the zine available as a pdf for anyone interested. As CSCW research related to environmental and climate justice is a new and emerging development, we hope that through this workshop, we can foster a longer-standing relationship with workshop participants engaging with these topics. We hope to sustain this connection through our Discord channel. If there is collective interest, we may establish a working group to discuss our work, and how we can support one another.

7 ORGANIZERS

Olivia Doggett: Olivia is a PhD student in the Faculty of Information and School of Environment at the University of Toronto. Her research investigates how migrant farm labor and agricultural technologies shape large-scale commercial farming practices and interrogates how these labor production systems can be redirected to support more sustainable and socially equitable agricultural practices.

Sarah Gram: Sarah is a PhD student at the Faculty of Information at the University of Toronto. Her research investigates repair and maintenance practices in socio-technical infrastructures, and is attentive to the material conditions that permit, inhibit, or encourage the examination of broken, old, or obsolete technologies. Her recent work has investigated labor practices in the field of consumer technology repair.

Jen Liu: Jen is a PhD student in Information Science at Cornell University (USA). Her research investigates the ecological, social, and political implications of computing technologies and infrastructures. Jen has worked on topics including land politics in digital agriculture and knowledge production in environmental sensing. Her current work is on the impact of climate change on networked infrastructures in the American South.

Ufuoma Ovienmhada: Ufuoma is a PhD student in Aerospace Engineering at MIT (USA). Her research investigates the opportunities, challenges, and limitations of satellite remote sensing for grassroots organizing in the Environmental Justice movement. Some of her recent work has investigated the uses of geospatial data for organizing at the intersection of prisons and the environment towards abolition.

Dr. Laura Perovich: Laura is an Assistant Professor in Art + Design at Northeastern University (USA). Her research aims to create impact on environmental challenges through local, community-engaged design practices that take seriously the complexities of social, cultural, and ecological systems and the possibilities of collective action. As part of this work, she designs experiences around making and making sense of environmental data in partnership

with communities, community groups, and environmental justice organizations.

Dr. Matt Ratto: Matt is a Professor in the Faculty of Information at the University of Toronto. His research focuses on how theories and perspectives from technoscience research can usefully extend and contextualize design and engineering practices. He studies and practices ‘critical making’, work that combines humanities insights and engineering practices and has published extensively on this concept.

Dr. Samar Sabie: Samar is an incoming Assistant Professor in the Institute of Culture, Communication, Information, and Technology and the Daniels Faculty of Architecture, Design, and Landscape at the University of Toronto. Her research examines how design as a socio-material practice supports communities in reconfiguring or reimagining the environments they inhabit in more equitable ways despite cultural, generational, and political differences. This research combines design and making with ethnography, critical theory, and pedagogy development in community-based contexts.

Dr. Robert Soden: Robert is an Assistant Professor at the University of Toronto working on climate informatics, human-centered computing, and science and technology studies. His research uses a range of ethnographic, participatory, and design research methods to evaluate and improve the technologies we use to understand and respond to environmental challenges like disasters and climate change.

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